Quiz 3 (Version 3)

CS 320: Concepts of Programming Languages

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- ▷ You will have approximately 35 minutes to complete this quiz. Make sure to read every question; some are easier than others.
- ▷ Make it very clear what your final solution for each problem is (e.g., by surrounding it in a box). We reserve the right to mark off points if we cannot tell what your final solution is.
- ▷ You must show your work on all problems unless otherwise specified. A solution without work will be considered incorrect (and will be investigated for potential academic misconduct).

1 OCaml Programming

(10 points) Implement the function

```
val map_filter2 : ('a -> 'b option) -> ('a -> 'c option) -> 'a list -> ('b * 'c) list
```

so that `map_filter2 f g l` is the list of pairs `(a, b)` such that `f x = Some a` and `g x = Some b` for some element `x` of `l`, in order.

```
let f x = if x mod 2 = 0 then Some (x / 2) else None
let g x = if x > 0 then Some (string_of_int x) else None
let _ = assert(map_filter2 f g [4;3;-2;6] = [(2,"4");(3,"6")])
let _ = assert(map_filter2 f g [1;3;5] = [])
let _ = assert(map_filter2 f g [-4;-2] = [])
let _ = assert(map_filter2 f g [] = [])
```

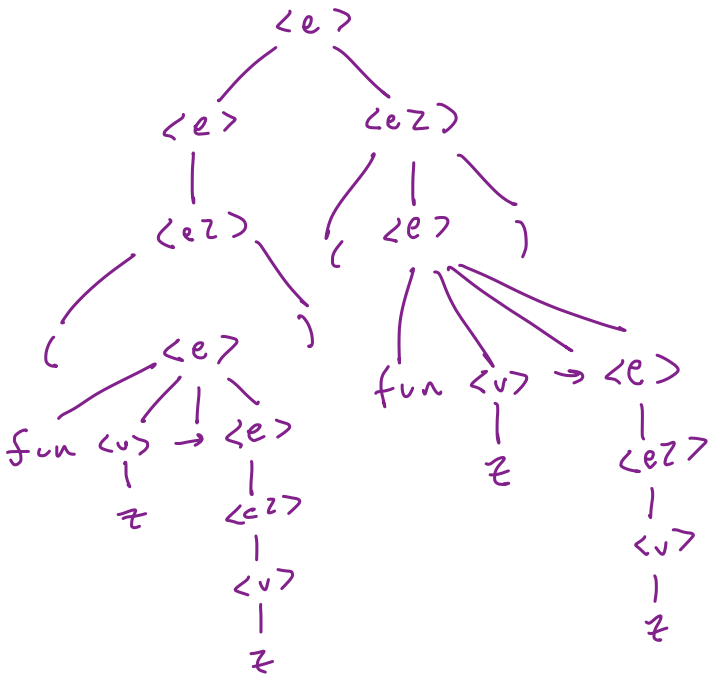
```
let map_filter2 f g l =
  let rec go l =
    match l with
    | [] -> []
    | x :: l ->
        match f x, g x with
        | Some b, Some c -> (b, c) :: go l r
        | _ -> go l r
  in go l
```

2 Formal Grammar

Consider the following BNF grammar.

```
<e>  ::= fun <v> -> <e> | <e> <e2> | <e2>
<e2> ::= <v> | (<e>)
<v>  ::= x | y | z
```

A. (10 points) Demonstrate that the sentence `(fun z -> z) (fun z -> z)` is recognized by the above grammar by providing a parse tree for it.



The grammar from the previous part is reproduced for convenience.

```
<e>  ::= fun <v> -> <e> | <e> <e2> | <e2>
<e2> ::= <v> | (<e>)
<v>  ::= x | y | z
```

B. (10 points) Demonstrate that the above grammar is ambiguous by determining a sentence with multiple parse trees. Include both parse trees in your answer.

fun x → x x

